coursera | UNIVERSITY OF LONDON

Applied Mathematical Methods for Computing

Module 1
Algebra, vectors and matrices

Module 2
Sequence and Series

Module 3
Combinatorics

Lesson 3.1 Counting

- Introduction
 Video • 1 min
- Counting
 Video • 5 min
- Complex counting
 Video • 4 min
- Counting
 Practice Assignment • Grade: 100%
- **Inclusion–Exclusion Principle**
 Dialogue • 30 min

3.2 The Pigeonhole Principle

- The Pigeonhole Principle
 Video • 3 min
- The Pigeonhole Principle – examples part 1
 Video • 4 min
- The Pigeonhole Principle – examples part 2
 Video • 3 min
- Use the Pigeonhole Principle
 Practice Assignment • 25 min
- Think of an example
 Discussion Prompt • 20 min
- Exercises with hints and tips
 Reading • 1h

Lesson 3.3 Order Matters

- Permutations
 Video • 6 min
- Permutations - solving a case study of your choice
 Practice Assignment • 30 min
- Permutations
 Practice Assignment • 20 min
- Combinations
 Video • 10 min
- No order/combinations
 Practice Assignment • 25 min
- Combinatorics
 Practice Assignment • 40 min

3.4 Summary and Assessments

- Exercises with hints and tips
 Reading • 10 min
- Conclusion
 Video • 3 min
- Check your understanding: End of module 3
 Graded Assignment • 20 min

Module 4
Probability and Statistics

Lesson 4.1 Probability Definition and Properties

- Introduction to probability
 Video • 11 min
- Probability
 Practice Assignment • 30 min

Lesson 4.2 Statistics

- Introducing statistics
 Video • 10 min
- Worked examples (socks from a drawer)
 Video • 7 min
- Sharing birthdays
 Video • 6 min
- Statistics
 Practice Assignment • 30 min

Lesson 4.3 Summary and Assessments

- Analysing Real-World Data
 Dialogue • 30 min
- Looking back
 Video • 2 min
- Check your understanding: End of module 4
 Graded Assignment • 20 min

Course Wrap-up

- Advanced Mathematical Methods for Computing: Course Summary
 Reading • 10 min
- The Ciphered Map: A Cartographic Escape Mystery
 Role Play • 15 min
- Course summary
 Video • 56 sec

Inclusion–Exclusion Principle

Welcome! In this Dialogue, we'll explore how to apply the inclusion-exclusion principle to solve complex counting problems involving overlapping sets. This is a crucial skill for many computer science applications!

Here's what we'll cover:

- **Problem Analysis and Information Organisation:** We'll break down a scenario, identify all relevant sets and their relationships, and organize the numerical data systematically.
- **Applying the Inclusion-Exclusion Formula:** We'll select and correctly apply the three-set inclusion-exclusion formula, making sure each component is properly identified and calculated.
- **Solution Verification and Interpretation:** We'll check our mathematical result for logical consistency and explain what the answer means in a real-world context.

Need help? Click "I'm stuck" at the top right to get a hint.

When you're ready, click "Start Dialogue"

Great! Let's dive into our first topic: **Problem Analysis and Information Organisation**. Imagine a professor is trying to select student assistants. They have 100 students in total. Out of these, 50 students are enrolled in the Algorithms class, 40 in the Data Structures class, and 30 in the Discrete Mathematics class.

To start, how would you begin to organize this information to prepare for applying the inclusion-exclusion principle?

S **Sandipan**

n(S)=100, n(A)=50, n(D)=40,n(M)=30

That's a good start by identifying the total number of students and the number of students in each individual class! You've correctly assigned variables to represent the size of each set.

Now, to fully prepare for the inclusion-exclusion principle, what other crucial pieces of information do we need to identify about these sets, especially concerning how they might overlap?

S **Sandipan**

S = AUDUM
we need n(A & D), n(A & M), n(D & M), n(A & D & M)

Excellent! You've hit on a very important point. We absolutely need to know about the overlaps between these classes to use the inclusion-exclusion principle effectively. You've correctly identified that we need the number of students in the intersections of two classes (like A and D, A and M, D and M) and the intersection of all three classes (A, D, and M).

Let's add some more details to our scenario:

- 15 students are enrolled in both Algorithms and Data Structures.
- 10 students are enrolled in both Algorithms and Discrete Mathematics.
- 5 students are enrolled in both Data Structures and Discrete Mathematics.
- 2 students are enrolled in all three classes (Algorithms, Data Structures, and Discrete Mathematics).

Now that we have all this information, how would you systematically list out all the known values for our sets and their intersections, using the notation you started with?

S **Sandipan**

n(AUDUM) = n(A)+n(D)+n(M)-n(A & D)-n(A & M)- n(D & M)+n(A & D & M) = 50 + 40 + 30 - 15 - 10 - 5 + 2

That's a fantastic step! You've not only listed the values but also jumped right into setting up the **Inclusion-Exclusion Formula** with those values. It's clear you understand which terms need to be added and which need to be subtracted.

Now, let's move on to our next topic: **Applying the Inclusion-Exclusion Formula**. Can you go ahead and calculate the result of the expression you've set up?

S **Sandipan**

92

You got it! The calculation results in 92.

Now that we've applied the formula and found a numerical answer, let's move on to our final topic: **Solution Verification and Interpretation**. What does this number, 92, actually represent in the context of our professor's student assistant selection problem?

S **Sandipan**

number of students that took at least one of the 3 classes

That's absolutely correct! The number 92 represents the total number of unique students who are enrolled in at least one of the three classes (Algorithms, Data Structures, or Discrete Mathematics). This is the size of the union of the three sets.

Now, to ensure our answer makes logical sense, how would you explain why the inclusion-exclusion principle is necessary here, rather than just simply adding up the number of students in each class?

S **Sandipan**

since we double count and triple count the students who are in the intersection regions

Precisely! You've hit on the core reason why the inclusion-exclusion principle is so powerful. If we simply added the numbers for each class, we would indeed be double-counting students who are in two classes and triple-counting those in all three. The principle systematically corrects for these overlaps.

Good job, you have completed all the topics for our session. Up next - view your feedback.

During today's session covering how to apply the inclusion-exclusion principle to solve complex counting problems, we focused on:

- **Problem Analysis and Information Organisation:** You successfully identified the individual sets and the crucial need to account for their intersections.
- **Applying the Inclusion-Exclusion Formula:** You correctly set up and calculated the formula for the given problem.
- **Solution Verification and Interpretation:** You accurately interpreted the result of the formula in the context of the problem and explained why the inclusion-exclusion principle is necessary.

Your strengths:

- You demonstrated a clear understanding of how to identify and organize the different components of a complex counting problem, including individual sets and their various intersections.
- You accurately applied the three-set inclusion-exclusion formula and correctly interpreted its result in the real-world context.

Areas for improvement:

- To further solidify your understanding, consider practicing more problems that require systematic data organization, perhaps using Venn diagrams, before directly applying the formula. This can help reinforce the visual representation of overlaps.

Keep up the good work!

The Dialogue has ended. Start a new chat to clear your chat history and start fresh.

Try again ✓ Completed

Dialogue is powered by AI, so check for mistakes and don't share any personal/sensitive information.

Like Dislike Report an issue Tell us what you think

Go to next item →